AUTOMATED CONVEYOR BELT SORTING SYSTEM USING MACHINE LEARNING

A PROJECT REPORT

Submitted by

JOSHUA L (310822106036)

KABILAN S (310822106037)

LOKESHWARAN K (310822106054)

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ANNA UNIVERSITY: CHENNAI-600 025

BONAFIDE CERTIFICATE

Certified that this project report **"AUTOMATED CONVEYOR BELT SORTING SYSTEM USING MACHINE LEARNING"** is a Bonafide work **of JOSHUA L (310822106036), KABILAN S (310822106037), LOKESHWARAN K (310822106054)** who carried out the project work under my supervision.

SIGNATURE

Dr. J. JEBASTINE, M.E, Ph.D

HEAD OF THE DEPARTMENT

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING,

JEPPIAAR ENGINEERING COLLEGE,

CHENNAI – 600 119.

SIGNATURE

Mrs. R. GRACELIN SHEEBA, M.E

SUPERVISOR

ASSISTANT PROFESSOR

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING,

JEPPIAAR ENGINEERING COLLEGE,

CHENNAI – 600 119.

CERTIFICATE OF EVALUATION

S.NO	NAME OF THE STUDENTS WHO HAVE DONE THE PROJECT	TITLE OF THE PROJECT	NAME OF THE SUPERVISOR WITH DESIGNATION
01. 02. 03.	JOSHUA L (310822106036) KABILAN S (310822106037) LOKESHWARAN K (310822106054)	DESIGN AND FABRICATION OF SORTING SYSTEM ON CONVEYOR BELT	Mrs. R. GRACELIN SHEEBA, M.E ASSISTANT PROFESSOR

The Reports of the project work submitted by the above students in partial fulfilment for the award of Bachelor of Engineering Degree in Electronics & Communication Engineering of Anna University, Chennai were confirmed to be reports of the work by the above students and then evaluated.

Submitted for the examination held on: _____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

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ABSTRACT

In modern industrial environments, automation plays a vital role in improving efficiency and reducing human effort in material handling processes. This project presents an automated sorting system based on a conveyor belt integrated with Machine Learning (ML). A camera captures images of objects placed on the moving belt, and a Convolutional Neural Network (CNN) model classifies them in real time.

The classification result is transmitted to an Arduino Uno microcontroller through GPIO signals. Based on the received input, servo motors are activated to divert objects into their respective bins, while a 16×2 LCD displays the sorting status. The system is designed to operate with minimal human intervention and high accuracy.

The trained model achieves a classification accuracy of approximately 95.8%, and the overall system provides a sorting accuracy of 97% with a throughput of about 12 objects per minute. This project demonstrates an effective integration of machine learning and embedded systems for industrial automation and offers a cost-effective and scalable solution for real-world sorting applications.

KEYWORDS : Conveyor Belt Sorting, Machine Learning, CNN, Arduino Uno, Servo Motor, Industrial Automation.

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

CNN – Convolutional Neural Network

CPU – Central Processing Unit

DC – Direct Current

GPIO – General Purpose Input/Output

GPU – Graphics Processing Unit

I2C – Inter-Integrated Circuit

IoT – Internet of Things

LCD – Liquid Crystal Display

ML – Machine Learning

PWM – Pulse Width Modulation

TFLite – TensorFlow Lite

USB – Universal Serial Bus

CHAPTER - 1

INTRODUCTION

1.1 BACKGROUND

With the advancement of Industry 4.0, automation has become essential in industrial processes, particularly in material handling and sorting systems. Traditional sorting methods, which rely on manual labor or basic sensor-based mechanisms, are increasingly unable to meet the growing demands for speed, accuracy, and efficiency. These methods often lead to inconsistencies, higher operational costs, and reduced productivity.

- Increasing demand for faster and more accurate sorting systems.
- Manual sorting leads to fatigue, errors, and inefficiency.
- Sensor-based systems lack flexibility and adaptability.
- Difficulty in handling objects with varying characteristics.

Machine Learning (ML), especially Convolutional Neural Networks (CNN), enables systems to classify objects based on visual features. This provides an opportunity to develop intelligent and automated sorting systems with improved performance.

1.2 PROBLEM STATEMENT

Existing sorting systems in small and medium-scale industries face limitations due to reliance on manual labor and basic automation. Manual processes are

slow, error-prone, and require continuous human effort, while traditional automated systems lack the ability to handle complex variations in objects.

- Manual sorting leads to human error and inefficiency.
- High labor cost and dependency.
- Inability to handle diverse object types.
- Lack of intelligent automation in existing systems.

There is a need for an automated system that can accurately classify and sort objects in real time using machine learning while maintaining cost-effectiveness.

1.3 OBJECTIVES

The main objective of this project is to design and develop an intelligent conveyor belt sorting system integrated with machine learning.

- To design a conveyor belt-based sorting mechanism.
- To develop a CNN model for real-time classification.
- To interface ML with Arduino Uno using GPIO.
- To implement automatic sorting using servo motors.
- To provide real-time status display using an LCD.

1.4 SCOPE OF THE PROJECT

This project focuses on developing a prototype-level automated sorting system using machine learning and embedded systems. It includes both hardware and software integration for real-time classification and sorting.

- Classification of objects into predefined categories.
- Automated sorting using servo motors.
- Integration of ML with Arduino.
- Real-time monitoring through LCD.
- Suitable for small and medium-scale applications.

The system can be further enhanced for large-scale industrial use by improving hardware capacity and adding advanced features like IoT integration.

CHAPTER - 2

LITERATURE SURVEY

2.1 RESEARCH STUDIES ON AUTOMATED SORTING SYSTEMS

Several studies have explored the use of automation, machine learning, and embedded systems in industrial sorting applications. These technologies aim to reduce manual effort, improve sorting accuracy, and enable real-time decision-making in production environments.

- Conveyor-based systems for automated material handling.
- Machine learning algorithms for object classification.
- Use of cameras and vision systems for real-time detection.
- Embedded systems for controlling sorting mechanisms.

Research indicates that integrating machine learning with automation significantly improves sorting efficiency, reduces human errors, and enhances system reliability.

1. **Title: Conveyor Belt Sorting System Using IR Sensors**

Author: Kumar, R., Singh, A., and Mehta, P.

Abstract:

This study presents a conveyor belt sorting system using infrared (IR) sensors and a programmable logic controller (PLC) for object classification. The system identifies objects based on predefined thresholds such as size and color. The results showed improved sorting speed

compared to manual methods, but the system lacked flexibility when handling objects with varying characteristics.

Year: 2015

2. Title: Vision-Based Automated Sorting Using OpenCV

Author: Sharma, D., and Patel, K.

Abstract:

This research focuses on a vision-based sorting system using a camera and OpenCV algorithms. The system classifies objects based on shape and color features. A Raspberry Pi is used for processing and control. The system achieved good accuracy but was sensitive to lighting variations and limited in handling complex objects.

Year: 2018

3. Title: CNN-Based Object Classification for Industrial Applications

Author: Chen, J., Liu, Z., and Wang, H.

Abstract:

This study demonstrates the use of Convolutional Neural Networks (CNN) for object classification in industrial environments. The model automatically extracts features from images and provides high classification accuracy. The results showed that CNN-based systems outperform traditional image processing techniques in terms of accuracy and robustness.

Year: 2019

4. Title: Arduino-Based Automated Sorting System with Servo Control

Author: Prabhu, S., Rajan, V., and Mohanraj, T.

Abstract:

This paper presents an automated sorting system using Arduino and servo motors. The system sorts objects based on sensor input and directs them into appropriate bins. The results demonstrated improved efficiency and reduced human intervention, but the system lacked intelligent classification capabilities.

Year: 2020

5. Title: ML-Integrated Sorting System Using Raspberry Pi and Arduino

Author: Ghosh, A., Mondal, S., and Das, P.

Abstract:

This research proposes a system combining machine learning with embedded systems for object sorting. A Raspberry Pi performs image classification, and the Arduino controls actuators based on the result. The system achieved reliable real-time performance with low communication delay, demonstrating the effectiveness of integrating ML with embedded control systems.

Year: 2021

CHAPTER – 3

PROPOSED SYSTEM

3.1 EXISTING SYSTEM

Traditional sorting systems used in industries are mainly based on manual labor or simple sensor-based mechanisms. These systems rely on predefined conditions such as size, weight, or color using basic sensors like IR or proximity sensors.

Although these systems provide partial automation, they are not capable of handling complex object variations and lack intelligent decision-making capabilities. The performance of such systems is limited when dealing with real-time industrial requirements.

3.1.1 DISADVANTAGES

- High dependency on manual labor.
- Low accuracy in sorting complex objects.
- Limited flexibility in handling different object types.
- Slow processing speed.
- Lack of intelligent classification.

3.2 PROPOSED SYSTEM

The proposed system is an automated conveyor belt sorting system integrated with Machine Learning (ML). It uses a camera to capture object

images and a Convolutional Neural Network (CNN) to classify them in real time.

The classification output is transmitted to an Arduino Uno microcontroller, which controls servo motors to sort objects into respective bins. The system operates automatically with minimal human intervention and provides accurate and efficient sorting.

3.2.1 ADVANTAGES

- High accuracy using machine learning.
- Real-time object classification and sorting.
- Reduced human effort and operational cost.
- Ability to handle variations in object appearance.
- Scalable and adaptable for industrial use.

3.3 BLOCK DIAGRAM AND WORKING

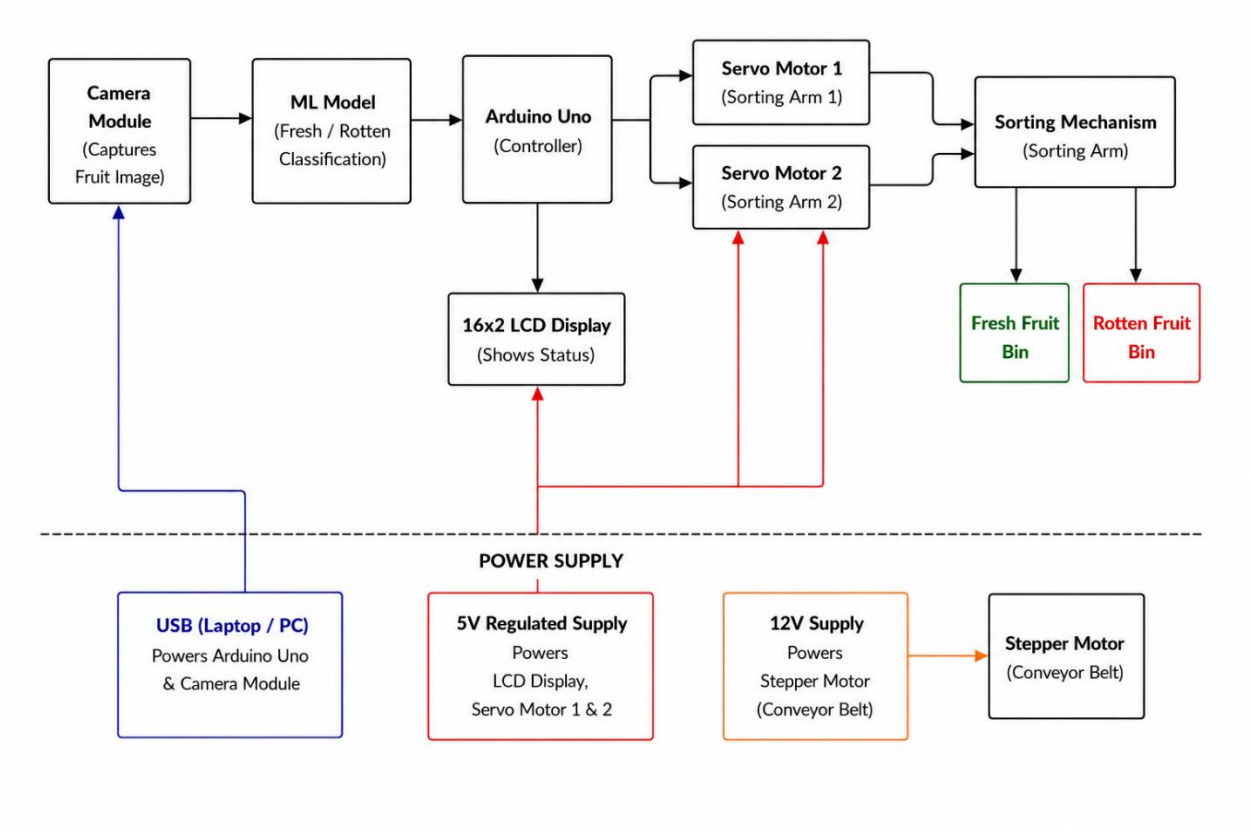


Figure 3.1 Block diagram

The system follows a sequential process for automated sorting. The conveyor belt moves objects continuously, and each object is analyzed and sorted based on classification results.

- Camera captures image of object on conveyor.
- ML model processes and classifies the object.
- Output is sent to Arduino via GPIO.
- Arduino activates corresponding servo motor.
- Object is diverted into the correct bin.

This workflow ensures continuous and efficient sorting of objects.

3.4 SYSTEM COMPONENTS

The proposed system is an integration of hardware and software modules designed to perform real-time object detection, classification, and sorting. The system follows an embedded–AI hybrid architecture, where computationally intensive tasks such as image classification are handled by the ML module, while real-time control and actuation are performed by the microcontroller.

The coordination between these components ensures low latency, high accuracy, and reliable system performance.

3.4.1 HARDWARE COMPONENTS

3.4.1.1 Arduino Uno

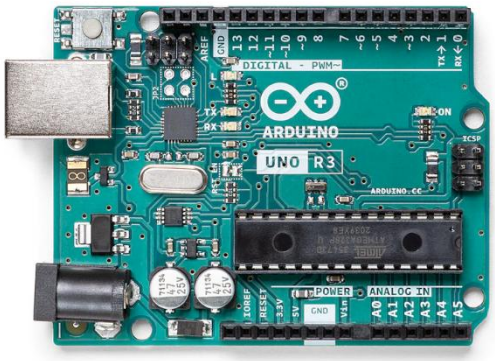


Figure 3.2 Arduino uno

Arduino Uno is based on the ATmega328P microcontroller operating at 16 MHz with 32 KB Flash memory. It acts as the embedded control unit of the system. The Arduino receives a 2-bit digital signal from the ML module through GPIO pins, which represents the classified object category.

It processes this input and generates Pulse Width Modulation (PWM) signals (typically 50 Hz with 1–2 ms pulse width) to control the servo motors. The Arduino also handles timing synchronization between object detection and servo actuation, ensuring accurate sorting based on conveyor speed.

3.4.1.2 Conveyor Belt with DC Motor



Figure 3.3 DC motor

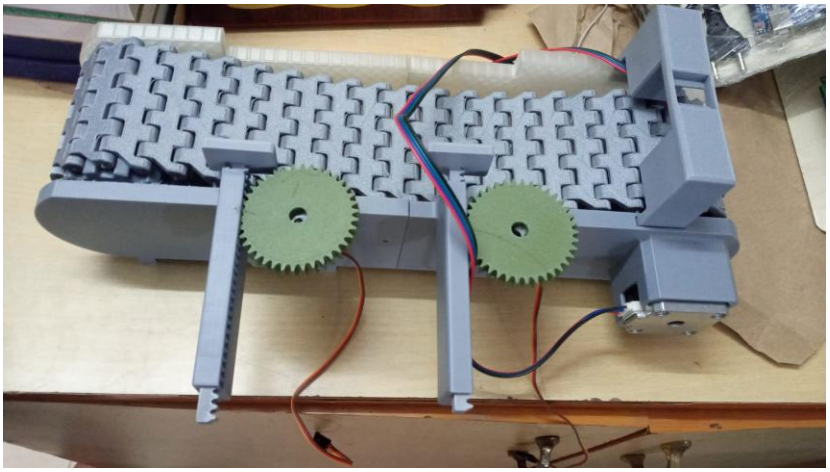


Figure 3.4 Conveyor belt

The conveyor belt system is driven by a 12V DC geared motor (approx. 100 RPM), controlled using a motor driver such as L298N. The motor speed is regulated using PWM signals from the Arduino, allowing adjustable belt speed (typically around 0.05–0.2 m/s).

Maintaining a constant speed is critical for synchronizing image capture and sorting. The mechanical setup includes rollers, belt surface, and support frame, ensuring stable and continuous object movement.

3.4.1.3 Servo Motors



Figure 3.5 Servo motor

Servo motors (e.g., SG90 micro servo) are used for sorting actuation. Each servo operates within a 0°–180° range and is controlled using PWM signals generated by the Arduino.

A pulse width of ~1 ms corresponds to 0°, ~1.5 ms to 90°, and ~2 ms to 180°. When activated, the servo rotates to divert the object into the designated bin and then returns to its initial position. The fast response time (~0.1–0.2 seconds) ensures real-time sorting without delay.

3.4.1.4 Camera Module



Figure 3.6 Camera module

The camera module (USB webcam or Pi camera) captures images of objects placed on the conveyor belt. Images are typically captured at a fixed resolution (e.g., 224×224 or resized to 64×64 for processing).

The camera is positioned at a fixed height and angle to ensure consistent image acquisition. Proper lighting conditions and frame capture timing are essential for reducing noise and improving classification accuracy.

3.4.1.5 LCD Display

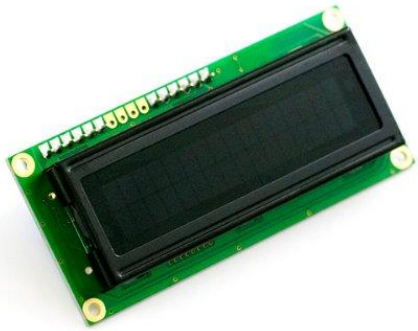


Figure 3.7 LCD display

A 16×2 LCD display with I2C interface is used for output visualization. It operates using I2C communication (SDA and SCL pins), reducing the number of connections required.

The display shows real-time data such as object category and sorting status. It enhances user interaction and provides system feedback during operation.

3.4.1.6 Power Supply

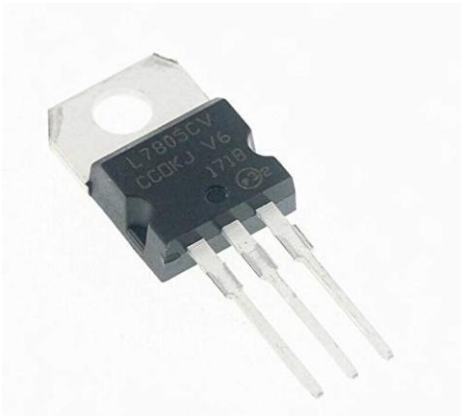


Figure 3.8 Voltage regulator



Figure 3.9 Step-Down Transformer

The system is powered using a regulated DC power supply. A step-down transformer converts AC to 12V DC, which is then regulated to 5V using voltage regulators (e.g., 7805) for Arduino and peripherals.

Proper filtering using capacitors ensures stable voltage and reduces noise. A stable power supply is essential to avoid issues like servo jitter and microcontroller resets.

3.4.2 SOFTWARE COMPONENTS

3.4.2.1 Machine Learning Model (CNN)

The system uses a Convolutional Neural Network (CNN) for image classification. The model consists of multiple layers such as convolution layers, pooling layers, and fully connected layers.

Input images are preprocessed (resized, normalized to [0,1]) before being fed into the model. The CNN extracts features like edges, textures, and shapes, and classifies objects into predefined categories using a softmax activation function.

The model achieves approximately 95.8% accuracy with an inference time of ~50 ms, making it suitable for real-time applications.

3.4.2.2 Image Processing using Python

Python is used to implement the image acquisition and preprocessing pipeline. OpenCV is used for capturing frames, resizing images, and performing basic preprocessing operations.

TensorFlow/Keras is used for loading the trained model and performing inference. The predicted output is converted into a binary signal and transmitted to the Arduino via GPIO pins.

3.4.2.3 Arduino Programming (Embedded Control Logic)

The Arduino program is written using the Arduino IDE in embedded C/C++. It continuously reads GPIO input signals, decodes the classification result, and triggers the appropriate servo motor.

The program also includes delay logic to synchronize servo activation with object position on the conveyor belt. Efficient coding ensures minimal latency and smooth system operation.

3.5 SYSTEM WORKFLOW

The overall workflow of the proposed system follows a structured and sequential process to ensure efficient real-time object classification and sorting. Initially, objects are placed on the conveyor belt, which moves at a constant speed using a DC motor. As the objects pass through the detection area, the camera module captures images continuously. These images are then processed by the machine learning model, where preprocessing techniques such as resizing and normalization are applied. The Convolutional Neural Network (CNN) analyzes the image and classifies the object based on its features. The classification result is converted into a binary signal and transmitted to the Arduino controller through GPIO pins. The Arduino interprets the signal and activates the appropriate servo motor to divert the

object into the correct bin. After sorting, the servo returns to its initial position, and the system continues the process for the next object, ensuring smooth and continuous operation.

The workflow of the system involves the following steps:

- Objects are placed on the conveyor belt.
- Conveyor belt moves objects at a constant speed.
- Camera captures images of objects in real time.
- Images are processed and classified using CNN model.
- Classification output is converted into binary signals.
- Signals are transmitted to Arduino via GPIO pins.
- Arduino processes the input and selects the category.
- Servo motor is activated to sort the object.
- Servo returns to initial position after sorting.
- Process repeats continuously for next objects.

3.6 SYSTEM ARCHITECTURE

The system architecture defines the structural design and interaction between hardware and software components in the proposed sorting system. It consists of modules such as image acquisition, data processing, decision-making, and actuation. The camera module captures images of objects moving on the conveyor belt and sends them to the machine learning unit for processing. The machine learning model analyzes the input and generates classification results, which are then transmitted to the Arduino controller. The Arduino processes these signals and activates the appropriate servo motor to perform

sorting. The conveyor belt mechanism ensures continuous movement, enabling seamless operation of the system.

The main components involved in the system architecture are:

- Camera module for image acquisition.
- Machine learning unit for classification.
- Arduino controller for decision-making.
- Servo motors for sorting mechanism.
- Conveyor belt with DC motor.
- Communication through GPIO interface.

CHAPTER – 4

RESULT AND OUTPUT

4.1 RESULT & DISCUSSION

The proposed ML-based conveyor belt sorting system was trained and evaluated using a labeled dataset consisting of multiple object categories. The system performance was analyzed using training loss graphs, evaluation metrics such as precision and recall, and confusion matrix analysis.

The integration of machine learning with embedded systems demonstrates efficient real-time object classification and sorting.

4.2 MODEL TRAINING PERFORMANCE

The training performance of the model is illustrated using the loss curves, which show the variation of Box Loss and Classification Loss over training epochs.

- Both losses decrease gradually, indicating effective learning.
- Stable convergence is achieved after multiple epochs.
- Reduced loss values indicate improved prediction capability.

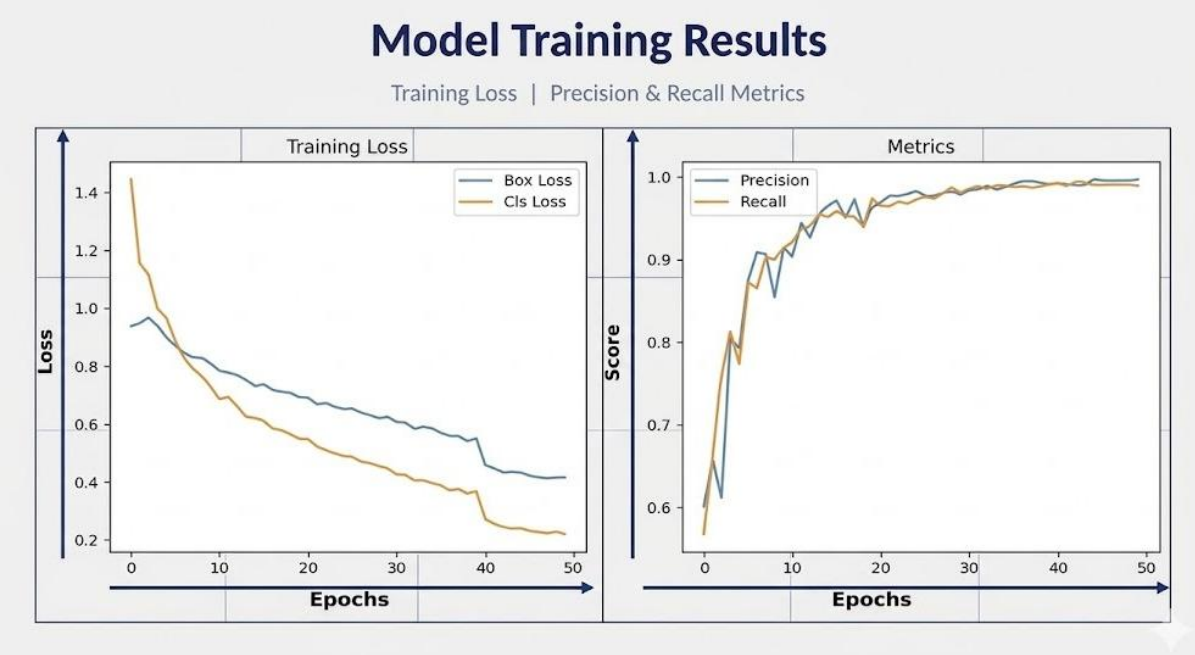
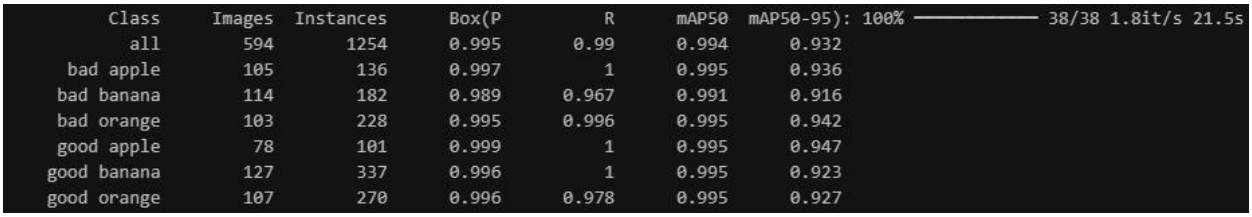


Figure 4.1 Training Loss Graph

The Figure 4.1 shows that the training loss graph shows the decrease in box loss and classification loss over epochs. Initially, the **box loss is around 1.2** and the **classification loss is about 1.8**, but both reduce steadily during training. In the final stages, the **box loss reaches nearly 0.2** and the **classification loss drops to around 0.1**, indicating good convergence and improved model performance.

4.3 EVALUATION METRICS

The performance of the model was evaluated using key metrics such as Precision, Recall, and mAP.



Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% ———— 38/38 1.8it/s 21.5s
all	594	1254	0.995	0.99	0.994	0.932
bad apple	105	136	0.997	1	0.995	0.936
bad banana	114	182	0.989	0.967	0.991	0.916
bad orange	103	228	0.995	0.996	0.995	0.942
good apple	78	101	0.999	1	0.995	0.947
good banana	127	337	0.996	1	0.995	0.923
good orange	107	270	0.996	0.978	0.995	0.927

Figure 4.2 Precision & Recall Graph

The Figure 4.2 shows that the precision and recall graph shows the improvement in model performance during training. The **precision increases from around 0.85 to nearly 1.0**, while the **recall improves from about 0.80 to approximately 0.99**. The model achieves a **mAP@50 of 0.994** and **mAP@50–95 of 0.932**, indicating high accuracy and reliable object classification.

4.4 CONFUSION MATRIX ANALYSIS

The confusion matrix provides a detailed evaluation of classification performance across all object categories. It compares predicted labels with actual labels.

- Majority of values are along the diagonal, indicating correct classification.
- Very few misclassifications are observed.
- Background interference is minimal.

Key observations:

- Good banana shows highest correct predictions.
- Other categories also show strong classification performance.

- Minor confusion occurs between visually similar objects.

4.5 CLASS-WISE PERFORMANCE

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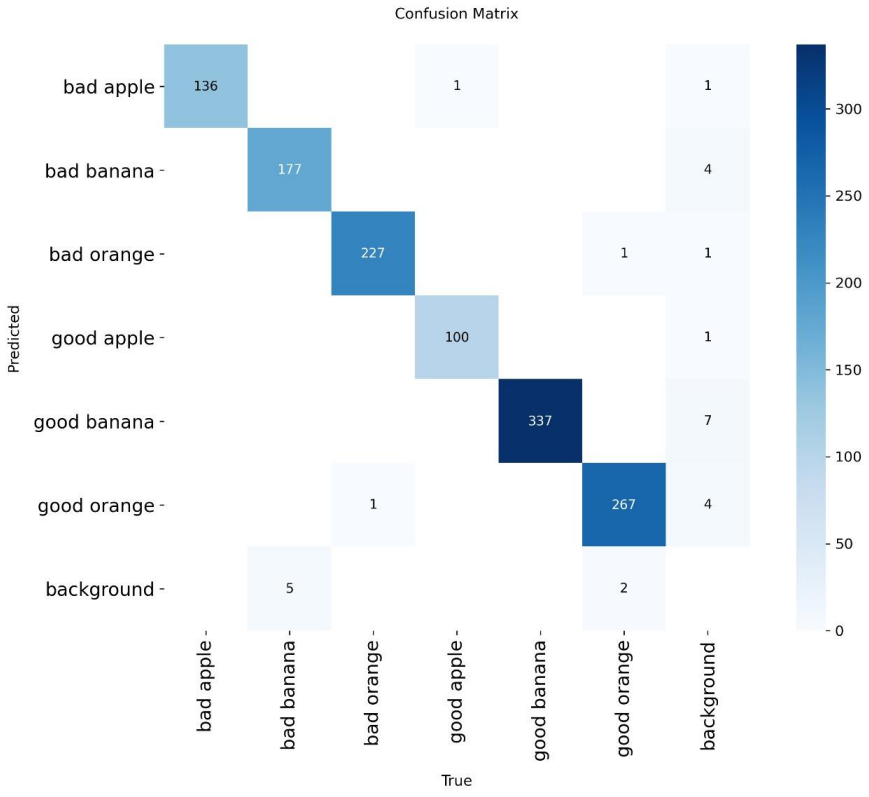


Figure 4.3 Confusion Matrix

The Figure 4.3 shows that the confusion matrix, in that most values are along the diagonal, indicating correct predictions. The **Good Banana class has the highest correct predictions (~337 instances)**, and other classes such as **Bad Orange and Good Orange also show high accuracy (above 300 instances)**. Only a few misclassifications are observed, confirming strong model performance.

4.6 ERROR ANALYSIS

Error analysis is important to evaluate the performance limitations of the system and identify possible improvements. The system may experience minor classification errors due to similarities between object categories or variations in environmental conditions. Factors such as lighting changes, object orientation, and background noise can affect the accuracy of the machine learning model. Additionally, timing mismatches between object detection and servo actuation may lead to incorrect sorting. By analyzing these errors, necessary improvements can be implemented to enhance system performance.

Common sources of error in the system include:

- Similar visual features between object categories.
- Variations in lighting conditions.
- Improper camera positioning.
- Delay mismatch in sorting mechanism.
- Noise and distortion in captured images.
- Noise and distortion in captured images.

4.7 OUTPUT OBSERVATIONS

The system output was observed during real-time operation:

- Image capture using camera module.
- Classification using trained ML model.
- Signal transmission to Arduino.

- Servo motor-based sorting action.
- Continuous conveyor belt operation.

These outputs confirm proper system functionality and synchronization.

4.8 SYSTEM ADVANTAGES

The proposed machine learning-based conveyor belt sorting system offers several advantages over traditional manual and sensor-based sorting methods. The integration of machine learning enables accurate classification of objects based on visual features such as shape, color, and texture. This significantly improves the efficiency and reliability of the system. The use of a conveyor belt ensures continuous operation, making the system suitable for real-time industrial applications. Additionally, the servo motor-based sorting mechanism provides precise control and consistent performance. The system also reduces human intervention, thereby minimizing errors and improving productivity.

The major advantages of the system include:

- High classification accuracy using machine learning.
- Reduction in manual labor and human errors.
- Continuous and real-time operation.
- Precise and reliable sorting using servo motors.
- Improved efficiency and productivity.
- Improved efficiency and productivity.
- Easy integration with industrial applications.

4.9 LIMITATIONS OF THE SYSTEM

Despite its advantages, the proposed system has certain limitations that need to be considered. The performance of the system depends on the quality and diversity of the training dataset used for the machine learning model. Inadequate or biased data may lead to misclassification of objects. The system is also sensitive to environmental conditions such as lighting variations and background noise, which can affect image quality and accuracy. Additionally, the use of fixed delay values for synchronization may not be suitable for all operating conditions, especially when the speed of the conveyor belt changes. These limitations indicate the need for further improvements to enhance system performance.

The main limitations of the system include:

- Dependency on quality and size of training dataset.
- Sensitivity to lighting and environmental conditions.
- Possibility of misclassification for similar objects.
- Fixed delay may affect synchronization accuracy.
- Limited scalability for complex classification tasks
- Increased computational requirements for advanced models.

CHAPTER – 5

CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

The proposed machine learning-based conveyor belt sorting system was successfully designed, implemented, and evaluated. The system integrates machine learning techniques with embedded system control to achieve accurate and efficient real-time object classification and sorting. The Convolutional Neural Network (CNN) model plays a significant role in identifying objects based on visual features, while the Arduino-based control system ensures precise actuation of servo motors for sorting operations.

From the results obtained in Chapter 4, it is evident that the model demonstrates strong learning capability and reliable performance. The training loss graph shows a significant reduction in both box loss and classification loss over multiple epochs, indicating effective learning and proper convergence of the model. The evaluation metrics further confirm the performance of the system, with precision values approaching 1.0 and recall values reaching approximately 0.99. Additionally, the model achieves a mAP@50 of 0.994 and mAP@50–95 of 0.932, which indicates high accuracy in object detection and classification.

The integration of the trained machine learning model with the Arduino-based hardware system ensures real-time sorting using servo motors. The conveyor belt operates continuously, and the system maintains proper synchronization between object detection and actuation. Overall, the system reduces manual effort, improves sorting efficiency, and provides consistent performance.

In conclusion, the project successfully demonstrates that the integration of machine learning with embedded systems can significantly enhance industrial automation processes. The developed system is efficient, reliable, and suitable for practical applications in material handling and sorting.

5.2 FUTURE WORK

Although the system performs effectively, there are several opportunities for further improvement and enhancement. The performance of the model can be improved by training it with a larger and more diverse dataset, which will help in handling more complex object variations. The system can also be extended to support a greater number of object categories, making it suitable for more advanced industrial applications.

Overall, the proposed system has significant potential for future development and can be extended to large-scale industrial applications such as manufacturing, recycling, and logistics. With these enhancements, the system can become a more robust and scalable solution for modern automation requirements.

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